

*make it*

*use it*

*run and adapt it*



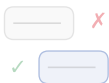
Pretrain

next token on  
~15T tokens



Scale

loss falls as  
a power law



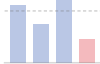
Post-train

base model  
to assistant



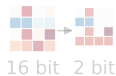
Prompt

steer it,  
no training



Evaluate

benchmarks  
and judges



Quantize

8B model:  
16 → 5 GB



LoRA

train < 1%  
of weights